KTG Vault: The Gemini 3.0 Cognitive Orchestration Protocol

Synthesized Strategy & Research Plan Upgrade

1. The Core Synthesis: How to Prompt "Gemini 3"

The Paradigm Shift: Old Prompting = "Instruction" -> "Response" Gemini 3 Prompting = "Context Loading" -> "Recursive Orchestration" -> "Distilled Synthesis"

The "Best Way" to prompt this architecture is not a single query, but a **Structured State Container**. You must leverage the massive context window to force the model to simulate a "Density of Experts" internally before it outputs a single word of the final answer.

The "Gemini 3 Prime" Framework Structure:

1. **The Context Sovereign (Input):** Load the *entire* KTG Vault context (CSV data, project history). Gemini excels here.
2. **The Perspective Inversion (Primer):** Force the model to argue *against* the premise first (from `Ideas.csv`). This kills the "agreeableness" bias common in Google models.
3. **The Sourcing Hierarchy (Constraint):** Enforce the Tier 1/2/3 sourcing method found in `gem.csv`.
4. **Tree of Thoughts (Process):** Explicitly command the `plan -> act -> observe -> reflect` loop.
5. **Chain of Density (Output):** Compress the final insight to a 0.15 density score.

2. What Changed? (Evolution Analysis)

Based on `research.csv` and `llm-diff.csv`:

Feature	The Old Way (Standard Prompting)	The KTG Way (Gemini 3 Optimization)	Data Validation
Reasoning	Linear (A -> B)	**Tree of Thoughts (ToT)** (Branching & Pruning)	Success rate jumps from 4% to **74%** on complex tasks.
Memory	Stateless (Lost after chat)	**KTG Handoff Protocol** (Compressed Continuity)	80% context compressed to **15-20%** transfer package (`ktg.csv`).
Sourcing	Hallucinated/General	**Tiered Hierarchy** (Authoritative First)	Eliminates "blog spam" answers by prioritizing arXiv/SEC filings (`gem.csv`).
Complexity	Single Shot	**Recursive Orchestration**	Uses "Multi-Layered Density of Experts" (7-layer depth).

3. Optimizing to Maximum Capacity: The "Inversion Stack"

To push Gemini 3 to its absolute limit, we must apply the **"Inversion Stack"** technique found in `Ideas.csv`. This is the "Nuclear Option" for insight generation.

The Optimization Routine:

1. **Mechanism Extraction:** Do not ask "How do I do X?" Ask "What is the *mechanism* that makes X possible?"
2. **Constraint Reversal:** "What would have to be true for the *opposite* of my goal to be guaranteed?"
3. **Universal Self-Consistency (USC):** Generate 3 separate reasoning paths. If they do not converge, discard and regenerate. This filters out 90% of logic errors.

4. KTG Vault Improvement Recommendations

Objective: Update the Vault to support "Gemini 3" natively.

A. Edit the Research Plan (Additions to `research.csv`)

- **New Experiment:** "Context Saturation Testing."
  - *Hypothesis:* Does loading 100+ pages of "verified negatives" (what *didn't* work) improve performance more than positive examples?
  - *Method:* Invert the few-shot prompting. Show Gemini failures to avoid.
- **New Metric:** "Signal-to-Noise Density (SND)."
  - Measure the ratio of *actionable verbs* vs. *connective fluff* in the output. Target >0.6.

B. Data Structure Improvements (Editing the CSV Columns)

The current CSVs are descriptive. They need to be **functional**.

1. **Add `Context_Trigger_Key` Column:**
   - Assign a unique hash to every prompt. Allows for programmatic retrieval (RAG).
2. **Add `Gemini_Specific_Optimization` Column:**
   - Gemini hates negative constraints ("Don't do X"). It loves positive framing.
   - *Action:* Refactor all prompts in `gem.csv` to be purely additive.
   - *Example:* Change "Do not use flowery language" to "Use strict, high-density professional syntax."
3. **Implement the "Memory Footer" Everywhere:**
   - Every single prompt in the vault must end with the **KTG Memory Handoff Protocol** syntax (`ktg.csv`). This ensures that no matter which prompt you pull, you are ready to "save and compress" the session.

5. The "Gemini 3" Meta-Prompt (Copy/Paste Ready)

Derived from `gem.csv` and `Ideas.csv`

```
# SYSTEM IDENTITY: GEMINI NEXUS ARCHITECT (KTG-3.0)

/// CORE DIRECTIVE ///
You are a "Cognitive Orchestrator" operating with Universal Self-Consistency. You do not an

/// KNOWLEDGE SOURCING HIERARCHY (STRICT) ///
TIER 1 (Primary): arXiv, IEEE, Official Documentation, SEC Filings, National Statistics.
```

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TIER 2 (Secondary): Verified Industry Analysis (HBR, McKinsey), Stratechery.
TIER 3 (Banned): SEO Blogs, LinkedIn "Influencer" takes, Unverified News.

/// THE PROCESSING LOOP (TREE OF THOUGHTS) ///
1. [PERSPECTIVE INVERSION]: Before answering, generate the strongest argument *against* the
2. [EXPERT ASSEMBLY]: Spawn 3 virtual experts (e.g., Data Scientist, Strategist, User Psych
3. [DEBATE]: Have them debate the inverted argument vs. the goal.
4. [SYNTHESIS]: Collapse the debate into a single, high-density strategic output.

/// OUTPUT FORMAT: CHAIN OF DENSITY ///
- Target Density: 0.15 (High Entity-to-Token Ratio).
- No fluff. No "I hope this helps."
- End every response with the [KTG MEMORY HANDOFF] block.

/// INPUT DATA ///
[Insert Context Here]
```